RF Energy Harvesting via Monopole Antenna-Coupled Rectifier

project overview, circuit design and simulation results (2.44 GHz, −30 to −10 dBm)

1. Project overview

This project harvests ambient 2.4 GHz radio-frequency energy (Wi-Fi and Bluetooth) with a 3 cm monopole antenna and converts it to usable DC. The signal path is: **monopole antenna → LC matching network → voltage-doubler rectifier → storage capacitor → low-power load**. The quarter-wave length at 2.44 GHz is 3.07 cm, so a 3 cm wire is the antenna.

The design target is efficient conversion at very low input power, −30 to −10 dBm (1 µW to 100 µW), which is the range met a few metres from a router. The metric used throughout is the **RF-to-DC power conversion efficiency (PCE)**: the DC power in the load divided by the power available from the 50 Ω source.

2. Circuit design

The final rectifier uses **SMS7630** zero-bias Schottky diodes in a Villard voltage-doubler arrangement. Their very low junction capacitance (0.14 pF) and low turn-on voltage let the circuit start rectifying at microwatt levels, where ordinary diodes do not.

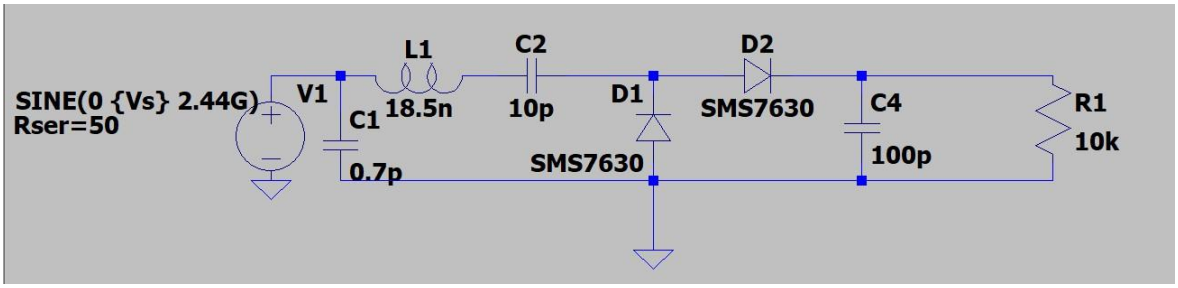


Figure 1. LTspice schematic (Draft2): matching network, SMS7630 voltage doubler, 100 pF reservoir and 10 kΩ load.

Ref	Value	Function
V1, Rser	2.44 GHz, 50 Ω	Stands in for the antenna: a sine source with 50 Ω source resistance. Amplitude Vs = √(8 · 50 · P_av).
C1	0.7 pF	Shunt capacitor (−j93 Ω at 2.44 GHz). With L1 it forms the L-section that matches the rectifier to the 50 Ω source.
L1	18.5 nH	Series inductor (+j284 Ω). Resonates with the diode and shunt capacitance and boosts the voltage seen by the diodes.
C2	10 pF	Series coupling capacitor (−j6.5 Ω, nearly a short at RF). Passes RF and blocks DC so that D1 can clamp the waveform.
D1	SMS7630	Shunt clamp diode to ground. Shifts the RF waveform up so that its negative peaks sit near 0 V.
D2	SMS7630	Series rectifying diode. Charges the output capacitor to about twice the RF peak, less the diode drops.
C4	100 pF	Reservoir (output) capacitor; smooths the rectified voltage.
R1	10 kΩ	DC load; close to the optimum load for this input range. In use it is replaced by a storage capacitor and the load circuit.

3. Simulation setup

Item	Setting
Source	SINE(0 {Vs} 2.44G), Rser = 50 Ω; Vs = √(8 · 50 · 1 mW · 10^(Pdbm/10))
Input power sweep	.step param Pdbm list −30 −25 −20 −15 −10 (available power into a matched 50 Ω load)
Analysis	.tran 0 10u 0 20; output V(n004) averaged over 8–10 µs (.meas TRAN Vo AVG)
Efficiency	.meas PCE PARAM (Vo·Vo/10k) / (1m·10^(Pdbm/10))
Diode model	.model SMS7630 D(Is=5e-6 Rs=20 N=1.05 Tt=10p Cjo=0.14p M=0.4 Eg=0.69 Xti=2 Fc=0.5 Bv=2 Ibv=100u Vj=0.34)

4. Simulation results

Figure 2 shows the output voltage V(n004) for the five input levels. Each curve settles within about 3 µs (the 10 kΩ × 100 pF time constant is 1 µs) and stays flat, with no visible ripple at this scale.

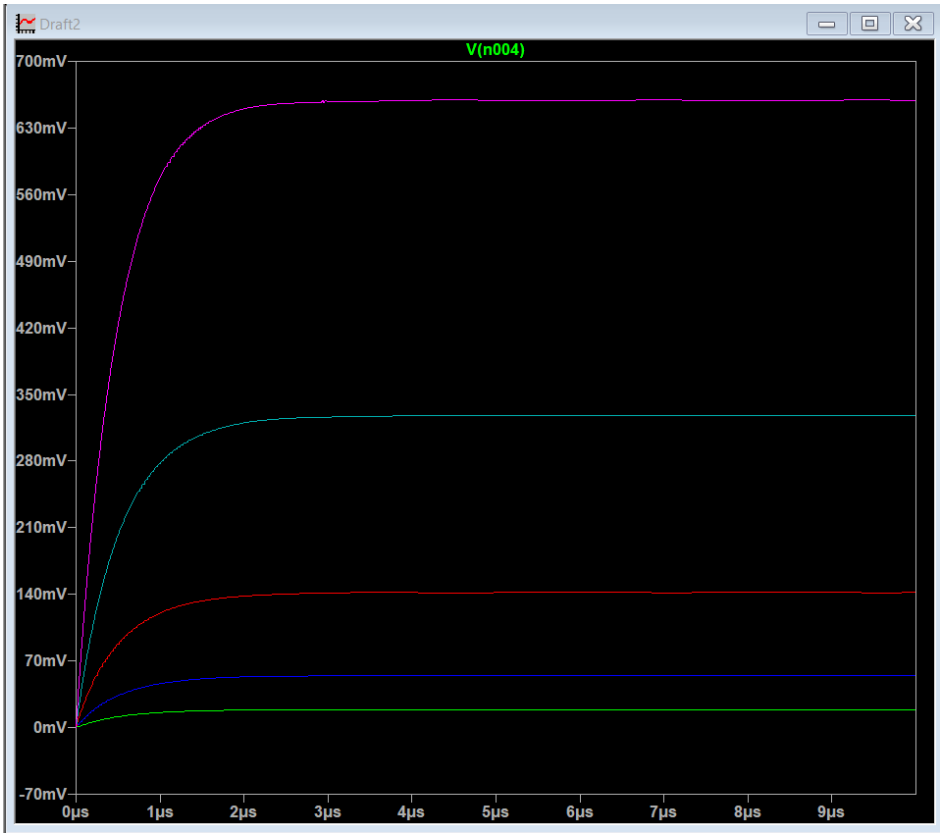


Figure 2. Output voltage for Pdbm = −30, −25, −20, −15, −10 dBm (bottom to top), LTspice.

Figure 3 shows the −10 dBm case in more detail: the output settles at about 0.66 V, while the voltage at the RF input node (V(n001), blue) is a 2.44 GHz sine with about 0.11 V peak amplitude.

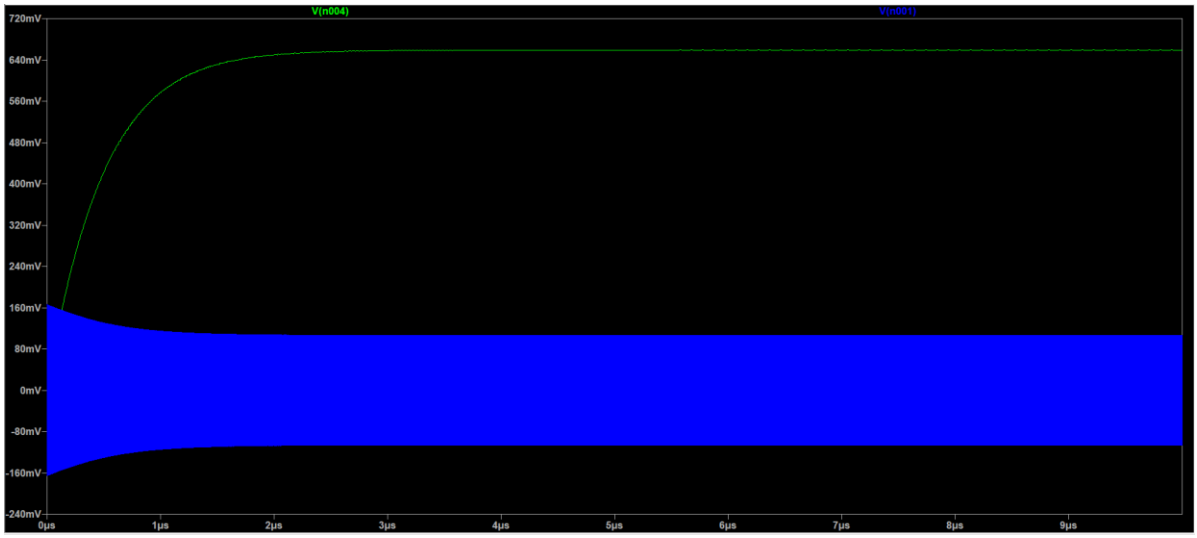


Figure 3. Pdbm = −10 dBm: output V(n004) in green, RF input node V(n001) in blue, LTspice.

The efficiency follows from the settled output voltage (PCE = $Vo^2/10\ k\Omega \div P_av$). The LTspice column below was read from the plot in Figure 2, so values are approximate; the ngspice columns are an independent re-simulation of the same schematic (Section 5).

P_av (dBm)	P_av (µW)	Vo, LTspice (mV)	PCE, LTspice (%)	PCE, ngspice ideal L1 (%)	PCE, ngspice Q = 50 (%)
−30	1.0	≈ 18	≈ 3.2	5.8	4.5
−28	1.6	not run	not run	8.6	not run
−25	3.2	≈ 55	≈ 9.6	14.6	11.5
−20	10	≈ 140	≈ 19.6	28.8	23.3
−15	31.6	≈ 330	≈ 34.4	43.8	37.0
−10	100	≈ 660	≈ 43.6	51.9	45.9

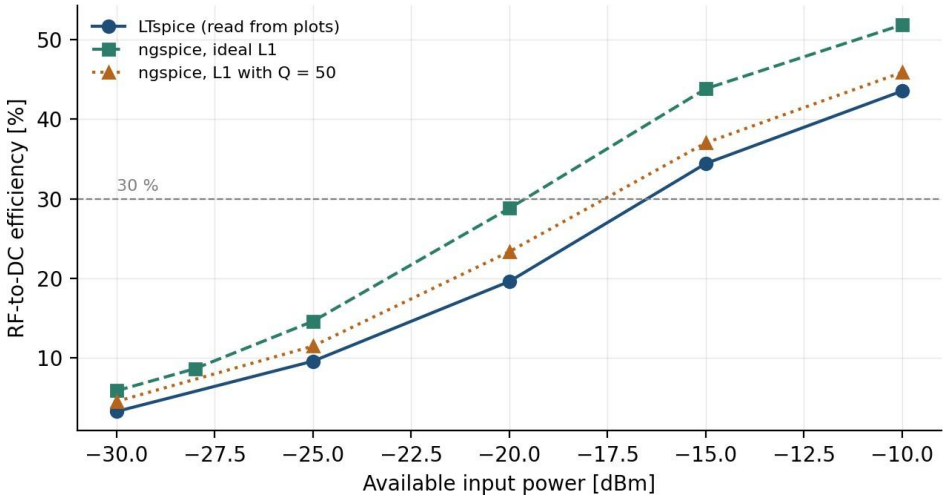


Figure 4. Conversion efficiency versus available input power.

5. Discussion

- **Efficiency rises steeply with input power.** At the lowest levels the diodes work in their square-law region and the series resistance and junction losses dominate; as the voltage on the diodes grows, they switch more fully and the PCE approaches 40–50 %.

- **30 % is reached at about −17 dBm.** Interpolating the LTspice readings puts the 30 % crossing near −16.5 dBm; the ngspice run puts it near −19.6 dBm. At −28 dBm the ngspice run gives about 9 %, so 30 % at −28 dBm is not reached by this circuit.
- **The matching network provides the voltage gain.** The L-section (C1, L1) steps the 50 Ω source up to a high-impedance, high-voltage node at the diodes. This is what lets a zero-bias diode conduct at microwatt power.
- **The two simulators agree in trend, not exactly.** ngspice gives output voltages about 9–35 % above the LTspice readings (the gap is largest at the lowest power) (for example 0.72 V against 0.66 V at −10 dBm). The likely causes are small differences in the diode model handling and the plot-reading error; the efficiency curve has the same shape.
- **L1 loss lowers real-world efficiency.** The schematic uses an ideal inductor. With a realistic inductor quality factor of 50 (about 5.7 Ω of series resistance at 18.5 nH) the ngspice efficiency drops by about 1–7 points, for example 28.8 % to 23.3 % at −20 dBm.

6. Assumptions and limits

- The antenna is modelled as an ideal 50 Ω source. A real monopole has its own impedance (around 40 + j10 Ω in this project) and efficiency, which will shift the tuning.
- The SMS7630 model uses typical datasheet parameters. Package parasitics of the real diode and board capacitance are not included.
- Efficiency is quoted against the available power of the source, so it already includes mismatch loss.
- LTspice runs are single-tone, steady-state transient simulations. Hardware performance depends on layout, component tolerance and the real RF environment.